

| Role        | Salary (UG)       | Salary (PG)       | Basis                   | Key Focus                     |
|-------------|-------------------|-------------------|-------------------------|-------------------------------|
| TCS Ninja   | Rs 3.36 LPA       | Rs 3.53 LPA       | Part A score only       | Aptitude + Verbal + Reasoning |
| TCS Digital | Rs 7.09-8.04 LPA  | Rs 7.09-8.04 LPA  | Part A + Part B         | Advanced Aptitude + Coding    |
| TCS Prime   | Rs 9.09-12.26 LPA | Rs 9.09-12.26 LPA | High Part B + Interview | DS&A + Coding + System Design |

## 1. Verbal Ability Syllabus

| Topic                      | Approx. Questions | Frequency (Repeated Years) |
|----------------------------|-------------------|----------------------------|
| Reading Comprehension      | 6 to 8            | 5 Stars — Asked every year |
| Sentence Correction        | 3 to 5            | 4 Stars — Very frequent    |
| Fill in the Blanks         | 3 to 5            | 4 Stars — Very frequent    |
| Error Detection / Spotting | 2 to 4            | 4 Stars — Very frequent    |
| Synonyms / Antonyms        | 2 to 3            | 3 Stars — Regular          |
| Para Jumbles / Sequencing  | 2 to 3            | 3 Stars — Regular          |
| One Word Substitution      | 1 to 2            | 3 Stars — Regular          |
| Idioms and Phrases         | 1 to 2            | 2 Stars — Occasional       |
| Vocabulary in Context      | 1 to 2            | 2 Stars — Occasional       |

## 2. Reasoning Ability Syllabus

| Topic                          | Approx. Questions | Frequency                  |
|--------------------------------|-------------------|----------------------------|
| Number / Letter Series         | 4 to 6            | 5 Stars — Asked every year |
| Blood Relations                | 3 to 5            | 5 Stars — Asked every year |
| Coding and Decoding            | 3 to 5            | 4 Stars — Very frequent    |
| Logical Puzzles / Arrangements | 4 to 6            | 4 Stars — Very frequent    |
| Syllogisms                     | 2 to 4            | 4 Stars — Very frequent    |
| Venn Diagrams                  | 2 to 3            | 3 Stars — Regular          |
| Statement and Conclusion       | 2 to 3            | 3 Stars — Regular          |
| Direction and Distance         | 1 to 3            | 3 Stars — Regular          |
| Visual / Spatial Reasoning     | 2 to 3            | 3 Stars — Regular          |
| Analogies (Word / Number)      | 1 to 2            | 2 Stars — Occasional       |
| Clocks and Calendars           | 1 to 2            | 2 Stars — Occasional       |

## 3. Numerical Ability Syllabus

|                                     |        |                            |
|-------------------------------------|--------|----------------------------|
| Time, Work and Pipes                | 3 to 5 | 5 Stars — Asked every year |
| Speed, Time and Distance / Trains   | 3 to 4 | 5 Stars — Asked every year |
| Percentages                         | 2 to 4 | 5 Stars — Asked every year |
| Profit, Loss and Discount           | 2 to 4 | 4 Stars — Very frequent    |
| Ratio, Proportion and Mixtures      | 2 to 3 | 4 Stars — Very frequent    |
| Simple and Compound Interest        | 2 to 3 | 4 Stars — Very frequent    |
| Data Interpretation (Tables/Charts) | 2 to 4 | 4 Stars — Very frequent    |
| Number System and HCF / LCM         | 2 to 3 | 4 Stars — Very frequent    |
| Permutation and Combination         | 1 to 3 | 3 Stars — Regular          |
| Probability                         | 1 to 2 | 3 Stars — Regular          |
| Geometry and Mensuration            | 1 to 2 | 2 Stars — Occasional       |
| Algebra / Linear Equations          | 1 to 2 | 2 Stars — Occasional       |

## Practice Questions

### 1. Numerical Ability

Q1. If the cost price of 15 articles equals the selling price of 12 articles, find the gain percent.

$$\text{Gain} = (15 - 12) / 12 \times 100 = 3/12 \times 100 = 25\%$$

✓ Answer: 25% | Repeated 4+ times across drives

Q2. A can do a piece of work in 12 days, B in 15 days. Working together, how many days will they take?

$$\text{A's 1-day work} = 1/12; \text{B's 1-day work} = 1/15 \text{ Together} = 1/12 + 1/15 = 5/60 + 4/60 = 9/60 = 3/20 \text{ Days} = 20/3 = 6.67 \text{ days (approximately 6 days 16 hours)}$$

✓ Answer: 6 and 2/3 days | Repeated 5+ times

Q3. A sum becomes Rs 1352 in 2 years at 4% per annum compound interest. Find the principal.

$$A = P(1 + r/100)^n \Rightarrow 1352 = P \times (1.04)^2 = P \times 1.0816 \quad P = 1352 / 1.0816 = \text{Rs } 1250$$

✓ Answer: Rs 1250 | Repeated 3+ times

Q4. A train 150 m long passes a pole in 15 seconds. How long to pass a 300 m platform?

Speed =  $150 / 15 = 10$  m/s Total distance =  $150 + 300 = 450$  m Time =  $450 / 10 = 45$  seconds

✓ Answer: 45 seconds | Repeated 6+ times

Q5. A shopkeeper marks goods 25% above cost price and gives a 10% discount. Find profit %.

Let CP = 100. MP = 125. SP =  $125 \times 0.90 = 112.5$  Profit = SP - CP =  $112.5 - 100 = 12.5\%$

✓ Answer: 12.5% | Repeated 4+ times

Q6. If 8 men can complete a job in 12 days, how many days will 6 men take for the same job?

Men x Days = constant  $\Rightarrow 8 \times 12 = 6 \times D$   $D = 96 / 6 = 16$  days

✓ Answer: 16 days | Standard work-men problem

Q7. A boat goes 30 km upstream in 6 hours and 30 km downstream in 3 hours. Find speed of boat in still water.

Upstream speed =  $30/6 = 5$  km/h; Downstream speed =  $30/3 = 10$  km/h Speed in still water =  $(10 + 5) / 2 = 7.5$  km/h

✓ Answer: 7.5 km/h | Repeated 3+ times

Q8. A sum doubles itself in 8 years at simple interest. Find the rate of interest per annum.

If P doubles, SI = P. SI =  $PRT/100 \Rightarrow P = P \times R \times 8 / 100$   $R = 100 / 8 = 12.5\%$  per annum ✓ Answer: 12.5%

Q9. Two pipes A and B can fill a tank in 20 and 30 minutes respectively. If both are opened together, in how many minutes will the tank be filled?

1-min work of A =  $1/20$ ; 1-min work of B =  $1/30$  Combined =  $1/20 + 1/30 = 3/60 + 2/60 = 5/60 = 1/12$  Tank fills in 12 minutes

✓ Answer: 12 minutes | Very frequent — Pipes & Cisterns

Q10. Find HCF of 36 and 84.

$36 = 4 \times 9 = 2^2 \times 3^2$ ;  $84 = 4 \times 21 = 2^2 \times 3 \times 7$  HCF =  $2^2 \times 3 = 12$

✓ Answer: 12

Q11. A car covers 300 km in 5 hours and a train covers 400 km in 8 hours. Find the ratio of their speeds.

Car speed =  $300/5 = 60$  km/h; Train speed =  $400/8 = 50$  km/h Ratio =  $60 : 50 = 6 : 5$

✓ Answer: 6:5

Q12. In how many ways can 5 boys be arranged in a row?

This is a permutation:  $5! = 5 \times 4 \times 3 \times 2 \times 1 = 120$

✓ Answer: 120 ways

Q13. What is the probability of getting a head when a fair coin is tossed?

Total outcomes = 2 (Head, Tail); Favourable = 1 (Head) Probability =  $1/2 = 0.5$

✓ Answer:  $1/2$  | Basic probability — always appears

Q14. The average of 5 numbers is 27. If one number is excluded, the average becomes 25. Find the excluded number.

Sum of 5 numbers =  $5 \times 27 = 135$  Sum of remaining 4 =  $4 \times 25 = 100$  Excluded number =  $135 - 100 = 35$

✓ Answer: 35 | Averages — very frequent

Q15. A 20% increase followed by a 20% decrease — what is the net change in value?

Net change =  $[20 + (-20) + (20 \times (-20))/100] \% = 0 - 4 = -4\%$  Net effect: 4% decrease

✓ Answer: 4% decrease | Classic trap question — Repeated 5+ times

Q16. The ratio of milk to water in a mixture is 3:1. If 8 litres of water is added, the ratio becomes 3:2. Find the original quantity of mixture.

Let milk =  $3x$ , water =  $x$ . After adding:  $3x / (x+8) = 3/2$   $6x = 3x + 24 \Rightarrow 3x = 24 \Rightarrow x = 8$  Original mixture =  $3(8) + 8 = 32$  litres

✓ Answer: 32 litres | Mixture & Alligation type

Q17. Find the next number in the series: 2, 6, 12, 20, 30, ?

Pattern:  $n(n+1)$  i.e.,  $1 \times 2, 2 \times 3, 3 \times 4, 4 \times 5, 5 \times 6 = 30$  Next =  $6 \times 7 = 42$

✓ Answer: 42 | Series — Repeated 6+ times

Q18. A man buys an item for Rs 800 and sells it for Rs 1000. Find the profit percentage.

Profit =  $1000 - 800 = \text{Rs } 200$  Profit % =  $(200/800) \times 100 = 25\%$

✓ Answer: 25%

Q19. Rs 5000 is invested at 10% per annum compound interest for 2 years. Find the amount.

$$A = P(1 + r/100)^n = 5000 \times (1.1)^2 = 5000 \times 1.21 = \text{Rs } 6050$$

✓ Answer: Rs 6050

Q20. Two numbers are in ratio 4:5. Their LCM is 80. Find the numbers.

Let numbers =  $4k$  and  $5k$ .  $\text{LCM} = 20k = 80 \Rightarrow k = 4$  Numbers = 16 and 20

✓ Answer: 16 and 20

Q21. A tank can be emptied by pipe C in 12 hours. Pipes A and B fill it in 4 and 6 hours. All three open together — in how long is tank filled?

Fill rate =  $1/4 + 1/6 - 1/12 = 3/12 + 2/12 - 1/12 = 4/12 = 1/3$  Tank fills in 3 hours

✓ Answer: 3 hours

Q22. A 100 m long train crosses a man in 10 seconds. Find its speed in km/h.

$$\text{Speed} = 100/10 = 10 \text{ m/s} = 10 \times (18/5) = 36 \text{ km/h}$$

✓ Answer: 36 km/h | Unit conversion — always asked

Q23. How many integers between 1 and 100 are divisible by both 2 and 3?

Divisible by both 2 and 3 means divisible by  $\text{LCM}(2,3) = 6$  Numbers: 6, 12, 18...

$$96 \Rightarrow \text{floor}(100/6) = 16 \text{ numbers}$$

✓ Answer: 16 | Number system type

Q24. Find the area of a circle whose circumference is 44 cm. (Use  $\pi = 22/7$ )

$$\text{Circumference} = 2\pi r = 44 \Rightarrow r = 44 \times 7 / (2 \times 22) = 7 \text{ cm} \text{ Area} = \pi r^2 = (22/7) \times 49 = 154 \text{ sq. cm}$$

✓ Answer: 154 sq. cm | Mensuration

Q25. What percent of  $2/3$  is  $1/6$ ?

$$\text{Required \%} = (1/6) / (2/3) \times 100 = (1/6 \times 3/2) \times 100 = (1/4) \times 100 = 25\%$$

✓ Answer: 25% | Fractions & Percentages

## 2. Reasoning Ability

Q1. If A is the son of B, B is the daughter of C, C is the father of D, how is A related to D?

B is daughter of C, so C is B's father. D is also a child of C, so B and D are siblings. A is son of B, so A is nephew/niece of D.

✓ Answer: A is D's Nephew

Q2. Find the odd one out: 8, 27, 64, 100, 125

$8 = 2^3$ ,  $27 = 3^3$ ,  $64 = 4^3$ ,  $125 = 5^3$  — all perfect cubes.  $100 = 10^2$  — NOT a perfect cube.

✓ Answer: 100 | Repeated 5+ times

Q3. Pointing to a photograph, a man says 'She is the daughter of my grandfather's only son.' How is the person in the photo related to the man?

Grandfather's only son = the man's father (assuming he has no uncles).  
Daughter of his father = his sister.

✓ Answer: Sister

Q4. Complete the series: A, C, F, J, O, ?

Gaps: +2, +3, +4, +5, +6... O is the 15th letter. Next =  $15 + 6 = 21$ st letter = U

✓ Answer: U | Letter series — Very frequent

Q5. In a row of 40 students, Ravi is 11th from the left. Priya is 13th from the right. How many students are between them?

Priya's position from left =  $40 - 13 + 1 = 28$  Students between =  $28 - 11 - 1 = 16$

✓ Answer: 16 | Ranking / Seating type

Q6. If DELHI is coded as 73541, and CALCUTTA as 82589662, how is CALICUT coded?

C=8, A=2, L=5, I=4, C=8, U=6, T=9 CALICUT = 8254869

✓ Answer: 8254869 | Coding-Decoding — Repeated 4+ times

Q7. Statements: All dogs are cats. Some cats are parrots. Conclusion: Some dogs are parrots

From 'All dogs are cats' and 'Some cats are parrots', we cannot confirm all dogs are in the parrot subset. The conclusion does NOT necessarily follow.

✓ Answer: Conclusion does not follow | Syllogism type

Q8. Complete: 3, 7, 13, 21, 31, ?

Differences: 4, 6, 8, 10, 12 (increasing by 2) Next term =  $31 + 12 = 43$

✓ Answer: 43

Q9. If Monday is coded as 2, Tuesday as 3, ... Sunday as 8, what is Friday coded as?

Monday=2, Tuesday=3, Wednesday=4, Thursday=5, Friday=6

✓ Answer: 6 | Coding-Decoding

Q10. A and B walk towards each other. A walks at 4 km/h from X and B at 6 km/h from Y. Distance XY = 100 km. When do they meet?

Relative speed =  $4 + 6 = 10$  km/h Time =  $100 / 10 = 10$  hours

✓ Answer: After 10 hours | Direction + Distance

Q11. From the given Venn diagram data: Total=100, only A=20, only B=25, both A and B=15. Find only those not in A or B.

Not in A or B = Total - (only A + only B + both) =  $100 - (20+25+15) = 40$

✓ Answer: 40 | Venn Diagrams — Repeated often

Q12. P is taller than Q but shorter than R. R is not the tallest. S is taller than R. Who is the shortest?

Order:  $Q < P < R < S$ . Since Q is the lowest in this chain, Q is shortest.

✓ Answer: Q

Q13. If RAIN is coded as UCLA, how is COLD coded?

$R+3=U$ ,  $A+2=C$ ,  $I+3=L$ ,  $N+1=O$  — different shifts. Apply same logic to COLD.  
 $C+3=F$ ,  $O+2=Q$ ,  $L+3=O$ ,  $D+1=E \Rightarrow FQOE$

✓ Answer: FQOE | Variable shift coding type

Q14. Three persons A, B, C sit in a row. A does not sit at the extreme ends. How many arrangements are possible?

A sits only in the middle position. B and C can be in  $2! = 2$  ways. Total = 2

✓ Answer: 2 arrangements

Q15. A clock shows 3:15. What is the angle between hour and minute hands?

At 3:00, angle = 90 degrees. In 15 min, hour hand moves  $15 \times 0.5 = 7.5$  degrees. Minute hand at 15 min = 90 degrees from 12. Angle =  $90 - 7.5 = 82.5$  degrees

✓ Answer: 7.5 degrees | Clocks — Repeated 3+ times

Q16. Complete: 1, 1, 2, 3, 5, 8, 13, ?

Fibonacci series — each term = sum of previous two.  $13 + 8 = 21$

✓ Answer: 21 | Classic series

Q17. Statements: No cat is a dog. All dogs are animals. Conclusion: No cat is an animal.

Even though no cat is a dog, cats could still be animals via another path. The conclusion is FALSE — it does not follow.

✓ Answer: Does not follow | Syllogism

Q18. If south-east becomes north, north-east becomes west, then what does south become?

SE→N (rotated 135 anticlockwise), NE→W (same 135 rotation). S rotated 135 anticlockwise = NE

✓ Answer: North-East | Direction Sense

Q19. Which number replaces the question mark: 4, 9, 25, 49, 121, ?

$2^2=4$ ,  $3^2=9$ ,  $5^2=25$ ,  $7^2=49$ ,  $11^2=121$  — squares of primes (2,3,5,7,11)

Next prime = 13,  $13^2 = 169$

✓ Answer: 169 | Squares of primes series

Q20. A man walks 5 km north, then 3 km east, then 5 km south. How far is he from the starting point?

North and south cancel (5 km each). Net displacement = 3 km east.

✓ Answer: 3 km east from starting point

Q21. In a certain code, MONKEY = XDJMKN. How is TIGER coded?

M→X (+11), O→D (-11), N→J (-4), K→M (+2)... pattern is complex. Each letter shifted in a specific pattern. Using same offsets: T→O, I→D, G→C, E→B, R→M ⇒ ODCBM

✓ Answer: ODCBM | Coding-Decoding

Q22. Six people are in a circle. A is between B and C. D is not next to A. E is next to D. F is between E and B. Who is next to C?

Arrangement: B-A-C-...-D-E-F-B (circular) Following the constraints: C is next to A and D

✓ Answer: A and D | Circular Arrangement



Q23. If 'white' means 'black', 'black' means 'red', 'red' means 'yellow', 'yellow' means 'blue', what is the colour of sky?

Sky is normally 'blue'. In this coding, 'blue' stands for... nothing changes for blue. Sky = blue = yellow (since yellow means blue in the code)

✓ Answer: Yellow | Colour coding — Repeated 4+ times

Q24. How many meaningful English words can be made from TPOSA using each letter only once?

TPOSA can form: TAPAS, PATOS, STOAP... Most valid = TAPOS, OATS, ATOP, TOPS, SPOT, STOP, POTS, OPTS Most TCS answers expect: 4 or 5 words (depends on exact letters given in exam)

✓ Answer: STOP, SPOT, POTS, TOPS, OPTS | Word formation

Q25. A is 3 ranks ahead of B in a class of 40. B's rank from last is 10. What is A's rank from the start?

B's rank from start =  $40 - 10 + 1 = 31$  A is 3 ranks ahead: A's rank =  $31 - 3 = 28$

✓ Answer: 28th from the start

### 3. Verbal Ability

Q1. Choose the correct option: "He \_\_\_\_\_ to the office every day despite the long distance."

Options: A) go B) goes C) going D) gone Subject = He (3rd person singular, present). Correct form = 'goes'.

✓ Answer: B) goes | Subject-Verb Agreement — Repeated 5+ times

Q2. Identify the error: 'She is one of the student who has passed the exam.'  
'One of the students' requires the plural form. 'student' should be 'students'.

✓ Answer: student => students | Error detection

Q3. Fill in the blank: 'The committee has submitted \_\_\_\_\_ report to the board.'  
Options: A) their B) his C) its D) our 'Committee' is a collective noun treated as singular => use 'its'.

✓ Answer: C) its | Pronoun agreement

Q4. Choose the synonym of LETHARGIC:

Options: A) Energetic B) Sluggish C) Alert D) Active Lethargic = lazy, slow, lacking energy. Synonym = Sluggish.

✓ Answer: B) Sluggish

Q5. Choose the antonym of VERBOSE:

Options: A) Talkative B) Wordy C) Concise D) Eloquent Verbose = using too many words. Antonym = Concise (using few words).

✓ Answer: C) Concise

Q6. Spot the error: 'Either the manager or the employees is responsible for this mistake.'

When using 'Either...or', the verb agrees with the nearer subject = 'employees' (plural). So 'is' should be 'are'.

✓ Answer: 'is' should be 'are' | Subject-Verb Agreement

Q7. Fill in the blank: 'I have been waiting here \_\_\_\_\_ two hours.'

Options: A) since B) for C) from D) until 'Since' is for a point in time; 'for' is for a duration. 'two hours' = duration => 'for'.

✓ Answer: B) for | Prepositions — Repeated 4+ times

Q8. Choose the correctly spelt word:

Options: A) Priviledge B) Privilege C) Privlege D) Privilage Correct spelling: P-R-I-V-I-L-E-G-E

✓ Answer: B) Privilege

Q9. Rearrange the jumbled sentence: 'books / she / reads / always / interesting' Correct order: She always reads interesting books. Subject + Adverb + Verb + Adjective + Object

✓ Answer: She always reads interesting books | Para jumble

Q10. Choose the word that best replaces the underlined phrase: 'He speaks in a very roundabout way without coming to the point.'

Options: A) Verbose B) Cryptic C) Circuitous D) Blunt Roundabout / not coming to the point = Circuitous (indirect)

✓ Answer: C) Circuitous | One-word substitution

Q11. Identify the correctly punctuated sentence:

Options: A) Its raining, outside. B) It's raining outside. C) Its' raining outside. D) Its raining outside. 'It's' = contraction of 'it is'. The sentence needs an apostrophe. ✓ Answer: B) It's raining outside.

Q12. Fill in the blank: '\_\_\_\_\_ he studied hard, he failed the exam.'

Options: A) Because B) Although C) Since D) Therefore The sentence shows contrast (studied hard but failed) => 'Although'

✓ Answer: B) Although | Conjunctions — Frequent

Q13. Choose the meaning of the idiom: 'Bite the bullet'

Options: A) Eat something hard B) Endure a painful situation C) Give up D) Fight hard 'Bite the bullet' = endure pain or hardship without complaining.

✓ Answer: B) Endure a painful situation

Q14. Select the correct passive voice: 'The teacher is teaching the students.'

Active: The teacher is teaching the students. Passive: The students are being taught by the teacher.

✓ Answer: The students are being taught by the teacher | Voice — Repeated 3+ times

Q15. Identify the correct reported speech: She said, 'I am going to the market.' Direct: 'I am going to the market.' Reported: She said that she was going to the market.

✓ Answer: She said that she was going to the market | Narration

Q16. Choose the synonym of BENEVOLENT:

Options: A) Cruel B) Generous C) Miserly D) Selfish Benevolent = well-meaning, kind, charitable => Generous

✓ Answer: B) Generous

Q17. Fill in the blank: 'The number of accidents \_\_\_\_\_ increased this year.'

Options: A) have B) has C) are D) were 'The number of' takes a singular verb => 'has'

✓ Answer: B) has | Tricky subject-verb agreement

Q18. Which sentence uses the article correctly?

Options: A) He is an honest man. B) He is a honest man. C) He is the honest man. D) He is honest man. 'Honest' starts with a vowel sound /o/, so use 'an' not 'a'.

✓ Answer: A) He is an honest man. | Articles — Repeated 5+ times

Q19. Choose the antonym of FRUGAL:

Options: A) Thrifty B) Economical C) Extravagant D) Careful Frugal = economical, not wasteful. Antonym = Extravagant (spending a lot).

✓ Answer: C) Extravagant

Q20. Spot the error: 'He had went to the market before she arrived.' 'Had went' is incorrect. Past perfect requires past participle: 'gone'. Correct: 'He had gone to the market before she arrived.'

✓ Answer: 'went' should be 'gone' | Tense error

Q21. Fill in: 'Neither the manager nor the employees \_\_\_\_\_ informed about the decision.'

When 'neither...nor' is used, verb agrees with the nearer subject = 'employees' (plural). So the blank = 'were'

✓ Answer: were | Correlative conjunctions

Q22. Choose the sentence with correct word usage:

A) She complemented him on his work.

B) She complimented him on his work.

'Complement' = complete/match. 'Compliment' = praise. Context needs praise => 'complimented'.

✓ Answer: B) She complimented him on his work. | Confusable words

Q23. Reading Comprehension tip — What is the best approach in TCS NQT RC? **1. Skim the passage in 60-90 seconds to get the main idea.**

**2. Read questions first, then find answers in the passage.**

**3. For 'Title' questions — pick the most general option that covers the whole passage.**

**4. Eliminate extreme/absolute options (words like 'always', 'never', 'only').**

✓ **Tip: RC questions are inference-based — the answer is in the passage, not your opinion**

Q24. Identify the meaning of: 'A penny saved is a penny earned.'

Options: A) Saving money is waste of time B) Being thrifty is as good as earning C) One must earn to save D) Penny is a small amount.

The proverb means: saving money has the same value as earning it.

✓ Answer: B) Being thrifty is as good as earning | Proverb/Idiom

Q25. Choose the option that best fills: 'The project was completed \_\_\_\_\_ schedule, surprising everyone.'

Options: A) in B) on C) ahead of D) behind 'Completed ahead of schedule' means finished before the deadline — which would surprise people.

✓ Answer: C) ahead of | Preposition in context

#### 4. Programming Logic MCQs

Q1. What is the output of: `print(type(5/2))` in Python 3?

In Python 3, '/' always performs true (float) division.  $5/2 = 2.5$ . `type(2.5) =`

✓ Answer: | Python basics — Repeated 5+ times

Q2. What is the time complexity of Binary Search?

Binary Search divides the array in half each step.  $T(n) = T(n/2) + O(1) \Rightarrow$

Time complexity =  $O(\log n)$

✓ Answer:  $O(\log n)$  | Algorithms — Always asked

Q3. Which data structure follows LIFO (Last In, First Out)?

Stack — the last element pushed is the first to be popped. Example: Function call stack, undo operations.

✓ Answer: Stack | Data Structures — Very frequent

Q4. What will be printed? `x = 10; y = 3; print(x % y) '%'` is modulus — gives remainder of x divided by y.

`10 % 3 = 1` (since `10 = 3x3 + 1`)

✓ Answer: 1 | Operators

Q5. What is the space complexity of Merge Sort?

Merge Sort requires auxiliary array for merging. Space complexity =  $O(n)$

✓ Answer:  $O(n)$  | Space complexity

Q6. Which sorting algorithm is stable?

Stable algorithms preserve the relative order of equal elements. Merge Sort and Bubble Sort are stable. Quick Sort is NOT stable by default.

✓ Answer: Merge Sort | Sorting properties — Repeated 3+ times

Q7. Output of: `for i in range(1, 10, 3): print(i)` `range(1, 10, 3)` generates: 1, 4, 7  
Prints: 1 then 4 then 7

✓ Answer: 1 4 7 | Python loops — Repeated 4+ times

Q8. Name the 4 pillars of OOP.

1. Encapsulation — binding data and methods together, hiding internal state.
2. Inheritance — child class acquires properties of parent class.
3. Polymorphism — same interface, different behavior (overloading/overriding).
4. Abstraction — hiding implementation, exposing only necessary features.

✓ Answer: Encapsulation, Inheritance, Polymorphism, Abstraction | Always asked

Q9. What is the difference between `'=='` and `'is'` in Python? `'=='` checks value equality: `5==5` is True.

`'is'` checks identity (same memory object): `a is b` checks if both point to same object.

✓ Answer: `'=='` compares values; `'is'` compares memory identity

Q10. What is the worst-case time complexity of Quick Sort?

Worst case occurs when the pivot is always the smallest or largest element.

Worst-case:  $O(n^2)$ . Average case:  $O(n \log n)$

✓ Answer:  $O(n^2)$  | Quick Sort — Repeated 4+ times

Q11. What does the `'break'` statement do in a loop?

'break' immediately terminates the loop, and control goes to the next statement after the loop. Compare: 'continue' skips the current iteration but does not exit the loop.

✓ Answer: Exits/terminates the loop immediately

Q12. What is a NULL pointer?

A pointer that does not point to any valid memory address. Dereferencing a NULL pointer causes a segmentation fault / runtime error.

✓ Answer: A pointer pointing to no valid address (value = 0 or NULL)

Q13. What is the output? `a = [1,2,3]; b = a; b.append(4); print(a)`

In Python, lists are mutable and assigned by reference. `b = a` means both `b` and `a` point to the SAME list. Appending to `b` also changes `a`. Output: `[1, 2, 3, 4]`

✓ Answer: `[1, 2, 3, 4]` | Python reference vs copy — Repeated 3+ times

Q14. What is the difference between a compiler and an interpreter? Compiler: translates entire source code to machine code at once before execution (e.g., C, C++). Interpreter: translates and executes code line by line (e.g., Python, JavaScript).

✓ Answer: Compiler = all at once; Interpreter = line by line

Q15. What is a recursive function?

Give a condition it must have.

A function that calls itself to solve a sub-problem. It MUST have a base case (termination condition) to prevent infinite recursion.

✓ Answer: A function calling itself + must have a base case

Q16. Time complexity of accessing an element in an array by index?

Arrays provide direct address computation:  $\text{address} = \text{base} + \text{index} * \text{size}$ .

Access time is constant regardless of array size. Time =  $O(1)$

✓ Answer:  $O(1)$  | Array complexity

Q17. What is the output? `print(bool(0), bool(""), bool([]), bool(1))` In Python, 0, empty string, and empty list are all Falsy. `bool(0)=False`, `bool("")=False`, `bool([])=False`, `bool(1)=True` Output: `False False False True`

✓ Answer: `False False False True` | Python truthy/falsy

Q18. What is an abstract class?

A class that cannot be instantiated directly. It can have abstract methods (no implementation) that must be overridden by subclasses. Used to define a template/interface for subclasses.

✓ Answer: A class with at least one abstract method that cannot be instantiated

Q19. What is the difference between ArrayList and LinkedList in Java?

ArrayList: backed by a dynamic array. Fast random access  $O(1)$ . Slow insert/delete in middle  $O(n)$ . LinkedList: doubly linked nodes. Slow random access  $O(n)$ . Fast insert/delete  $O(1)$  with iterator.

✓ Answer: ArrayList=fast access; LinkedList=fast insert/delete

Q20. What is a constructor in OOP?

A special method automatically called when an object is created. It has the same name as the class and no return type (in Java/C++). Purpose: initialise the object's attributes.

✓ Answer: Special method called at object creation to initialise attributes

Q21. What does DRY stand for in programming principles?

DRY = Don't Repeat Yourself. Every piece of logic should exist in exactly one place in the codebase. Reduces redundancy, makes code easier to maintain.

✓ Answer: Don't Repeat Yourself

Q22. What is the output of: `print(2**10)` in Python?

'\*\*' is the exponentiation operator in Python.  $2^{**}10 = 2^{10} = 1024$

✓ Answer: 1024 | Python operators

Q23. What is the difference between deep copy and shallow copy in Python?

Shallow copy: copies the object but still references the same inner objects.

Deep copy: copies the object AND all nested objects recursively. Use `copy.deepcopy()` for a complete independent clone.

✓ Answer: Shallow=same inner refs; Deep=fully independent copy

Q24. What is the time complexity of inserting an element at the beginning of a singly linked list? To insert at beginning: create new node, point it to current head, update head. This takes constant time regardless of list size. Time =  $O(1)$

✓ Answer:  $O(1)$  | Linked List



Q25. What does OOPS 'Polymorphism' mean? Give an example.

Polymorphism = many forms. Same method name behaves differently based on context. Method Overloading: same name, different parameters (compile-time). Method Overriding: child class redefines parent method (runtime).

Example: Animal.speak() => Dog: bark(), Cat: meow()

✓ Answer: Same method, different behaviour | Always asked in interviews too

## 5. CODING QUESTIONS

Q1 — Push Zeros to End (Array Manipulation)

A chocolate factory packs chocolates in packets. Empty packets are represented by 0 in an integer array of size N. Move all 0s to the end while maintaining the relative order of non-zero elements.

Q2 — Odd-Even Vehicle Fine (Simulation / Conditions)

Delhi government allows odd-numbered vehicles on odd dates and even-numbered on even dates. Given date D, number of vehicles N, and their last digits, and fine amount X, calculate total fine.

Example: Input: N=6, D=12, X=200, digits=[1,2,5,3,4,7] Output: 800 (vehicles 1,5,3,7 violated)

Q3 — Fibonacci Custom Sequence (Math / Series)

Print first N terms of a Fibonacci-like sequence where the first two terms are given as A and B (instead of 0 and 1). Each subsequent term = sum of two preceding terms.

Example: Input: A=3, B=5, N=7 Output: 3 5 8 13 21 34 55

Q4 — Check Prime and List Primes (Number Logic)

Given a number N, check if it is prime. If prime, print 'Prime'. If not, print 'Not Prime'. Either way, print all prime numbers from 2 to N.

Example: Input: N=20 Output: Not Prime | 2 3 5 7 11 13 17 19

Q5 — Reverse and Check Palindrome (String / Number)

Given a number N, print its reverse. Then check if the number is a palindrome. Print 'Palindrome' or 'Not Palindrome'.

Example: Input: 12321 Output: 12321 | Palindrome

Q6 — Maximum Subarray Sum (Array — Kadane's Algorithm)

Given an array of N integers (may include negatives), find the maximum sum of any contiguous subarray.

Example: Input: arr=[-2, 1, -3, 4, -1, 2, 1, -5, 4] Output: 6 (subarray: [4,-1,2,1])